Chapter 1: The Guitar and the Drum

Special Functions of Mathematical Physics: A Tourist's Guidebook

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June 12, 2026

Outline

Where we are going

Two instruments, one idea.

- A plucked **guitar string** and a struck **circular drum** are both described by the *wave equation*, which we build from a lattice of masses and springs (Hooke's law).
- **Separation of variables** turns each wave equation into ordinary differential equations for the spatial shape and the time evolution.
- For the string the spatial equation gives **sinusoids** (Fourier modes). For the drum it gives **Bessel's equation**, whose solutions are not built from elementary functions.
- Functions that arise this way but cannot be written in closed form with elementary functions are named: they are the **special functions** of the title.

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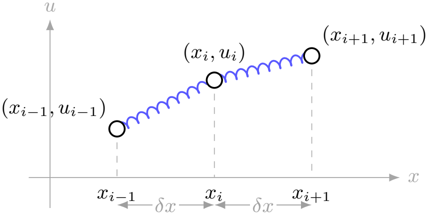
From masses and springs to the wave equation

We model a plucked string by its vertical displacement $u(x, t)$, a function of the horizontal position x along the string and the time t. Replace the continuous string by n small weights of mass m at nodes $\big(x_i,\ u(x_i, t)\big)$, joined to their neighbors by massless springs. The nodes move only vertically, so the horizontal positions x_i are fixed with uniform spacing δx.



From masses and springs to the wave equation (cont.)

Let $u_i = u(x_i, t)$ be the displacement of node i. Each spring pulls a node toward its neighbor with a force proportional to the relative displacement (Hooke's law, stiffness κ). Newton's second law for node i balances its acceleration against the pull from the spring above and the spring below:

$$F = m\,\frac{d^2 u_i}{dt^2} = \kappa\,(u_{i+1} - u_i) - \kappa\,(u_i - u_{i-1}). \tag{1.1.1}$$

The two terms are the two neighbors: the spring to node $i+1$ pulls up by $\kappa(u_{i+1} - u_i)$, and the spring to node $i-1$ pulls by $-\kappa(u_i - u_{i-1})$.
Now introduce the totals for a string of length $L = n\,\delta x$: total mass $M = nm$ and total stiffness $K = \kappa/n$. We rewrite the right-hand side of (1.1.1) to expose a difference quotient. Combining the two spring terms,

$$\kappa(u_{i+1} - u_i) - \kappa(u_i - u_{i-1}) = \kappa\,(u_{i+1} - 2u_i + u_{i-1}).$$

From masses and springs to the wave equation (cont.)

Divide (1.1.1) by m and multiply and divide by $(\delta x)^2$ so the bracket becomes a centered second difference. Multiplying and dividing by $(\delta x)^2$ leaves the value unchanged, $\frac{\kappa}{m} = \frac{\kappa}{m}(\delta x)^2 \cdot \frac{1}{(\delta x)^2}$, so naming the constant prefactor $c^2 := \frac{\kappa}{m}(\delta x)^2$,

$$\frac{d^2 u_i}{dt^2} = \frac{\kappa}{m}\left(u_{i+1} - 2u_i + u_{i-1}\right) = \frac{\kappa}{m}(\delta x)^2\,\frac{u_{i+1} - 2u_i + u_{i-1}}{(\delta x)^2} = c^2\,\frac{u_{i+1} - 2u_i + u_{i-1}}{(\delta x)^2}. \tag{1.1.2}$$

To see that this prefactor is finite and independent of n, use $\kappa = Kn$, $m = M/n$, and $L = n\,\delta x$ (so $\delta x = L/n$):

$$c^2 = \frac{\kappa}{m}(\delta x)^2 = \frac{Kn}{M/n}\left(\frac{L}{n}\right)^2 = \frac{Kn^2}{M}\cdot\frac{L^2}{n^2} = \frac{KL^2}{M},$$

so $c^2 = KL^2/M$: the powers of n cancel, leaving a fixed material constant. The centered difference $(u_{i+1} - 2u_i + u_{i-1})/(\delta x)^2$ is exactly the finite-difference approximation to the second spatial derivative. In the limit $\delta x \to 0$ (equivalently $n \to \infty$) it becomes $\partial^2 u/\partial x^2$, and (1.1.2) turns into

$$\frac{\partial^2 u}{\partial t^2} = c^2\,\frac{\partial^2 u}{\partial x^2}. \tag{1.1.3}$$

This is the (one-dimensional) *wave equation*, and c is the *wave speed*.

From masses and springs to the wave equation (cont.)

Equation (1.1.3) has two time derivatives and two space derivatives, so a unique solution needs two initial conditions and two boundary conditions. For a plucked string of length $L = 1$, clamped at both ends and released from rest in some initial shape $u_0(x)$:

$$\begin{aligned} \text{initial:} \quad & u(x,0) = u_0(x), \qquad && \frac{\partial}{\partial t} u(x,0) = 0, \\ \text{boundary:} \quad & u(0,t) = 0, && u(1,t) = 0. \end{aligned}$$

The string starts in the shape u_0 with zero velocity, and its endpoints never move.

Separation of variables for the string

We look for a solution that factors into a pure shape times a pure time signal,

$$u(x,t) = X(x)\,T(t). \tag{1.1.4}$$

Substituting (1.1.4) into the wave equation (1.1.3), and writing a prime for differentiation with respect to a function's *independent* variable, gives $XT'' = c^2\,TX''$. Dividing by XT separates the variables:

$$\frac{T''}{T} = c^2\,\frac{X''}{X}. \tag{1.1.5}$$

The left side depends only on t; the right side depends only on x. Two expressions in different variables can stay equal for all x and t only if both equal the same constant. Calling that constant λ,

$$\frac{1}{c^2}\,\frac{T''}{T} = \frac{X''}{X} = \lambda. \tag{1.1.6}$$

We test the sign of λ by writing $\lambda = \pm k^2$ for a real constant k.
Positive constant $\lambda = +k^2$. Then $X'' = k^2X$, with general solution

$$X(x) = A\,e^{kx} + B\,e^{-kx}.$$

Separation of variables for the string (cont.)

The clamped ends require $X(0) = X(1) = 0$. This is a homogeneous linear system for (A, B),

$$A + B = 0, \qquad Ae^{k} + Be^{-k} = 0,$$

whose coefficient determinant is $e^{-k} - e^{k} = -2\sinh k$. For any $k \neq 0$ this is nonzero, so the only solution is the trivial one: from $A = -B$ the second equation gives $B(e^{-k} - e^{k}) = 0$, hence $B = 0$ and then $A = 0$. Thus $X \equiv 0$: only the trivial (silent) string. We reject this case.

Zero constant $\lambda = 0$. Then $X'' = 0$, so $X(x) = Ax + B$. The boundary conditions give $B = 0$ from $X(0) = 0$ and then $A = 0$ from $X(1) = 0$. This also gives only the silent solution, so the zero mode is rejected.

Separation of variables for the string (cont.)

Negative constant $\lambda = -k^2$. Then

$$X'' = -k^2 X, \qquad T'' = -c^2 k^2 T.$$

The spatial equation has oscillatory solutions

$$X(x) = A \cos kx + B \sin kx.$$

Imposing $X(0) = 0$ kills the cosine, $A \equiv 0$. Imposing $X(1) = B \sin k = 0$ with $B \neq 0$ forces $\sin k = 0$, hence $k = n\pi$ for a positive integer $n = 1, 2, \ldots$. The value $n = 0$ is the zero mode already rejected, and negative n repeats the same sine shape up to a minus sign. Each admissible positive n gives one allowed half-wavelength on the string. The time equation $T'' = -c^2 k^2 T$ has solution

$$T(t) = C \cos ckt + D \sin ckt.$$

The release-from-rest condition $\partial_t u(x, 0) = 0$ means $T'(0) = 0$. Since $T'(0) = ckD$, we need $D \equiv 0$, leaving $T(t) = C \cos ckt$.

Separation of variables for the string (cont.)

Multiplying the surviving factors and writing $k = n\pi$, each positive integer n yields an independent standing-wave mode

$$u_n(x, t) = a_n \sin(n\pi x) \cos(cn\pi t), \tag{1.1.7}$$

with $a_n = BC$ a single constant per mode.

The Fourier expansion of the string

The wave equation is linear, so any sum of the modes (1.1.7) is again a solution. Superposing all of them,

$$u(x,t) = \sum_{n=1}^{\infty} a_n \sin(n\pi x) \cos(cn\pi t). \tag{1.1.8}$$

The coefficients a_n are fixed by the remaining condition, the initial shape. Setting $t = 0$ (where every cos equals 1) gives

$$u(x,0) = \sum_{n=1}^{\infty} a_n \sin(n\pi x) = u_0(x). \tag{1.1.9}$$

Expression (1.1.9) is the *Fourier expansion* of $u_0(x)$, and each a_n is a *Fourier coefficient*. The pure sinusoids $\sin(n\pi x)$ are the special functions of the guitar string—except that here they are elementary, so we already know them. The drum will not be so kind.

The Fourier expansion of the string (cont.)

To compute the coefficient a_m, multiply (1.1.9) by $\sin(m\pi x)$ and integrate:

$$\int_0^1 u_0(x)\sin(m\pi x)\,dx = \sum_{n=1}^{\infty} a_n \int_0^1 \sin(n\pi x)\sin(m\pi x)\,dx = \frac{a_m}{2}.$$

Therefore

$$a_m = 2\int_0^1 u_0(x)\sin(m\pi x)\,dx.$$

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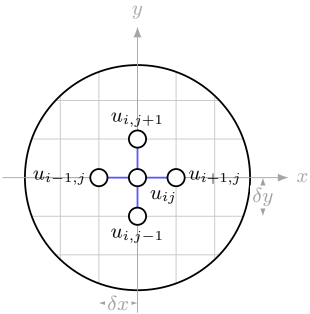
The two-dimensional wave equation

Now model a circular drum head. We again build the equations of motion from Hooke's law, but this time the masses form a two-dimensional lattice connected by massless springs in both the x and y directions. For the vertical displacement $u_{ij}(t) = u(x_i, y_j, t)$ of the node at (x_i, y_j), assume the mesh and spring scaling have been chosen isotropically so that the horizontal and vertical continuum limits share one wave speed c. Then the four neighbors (left/right and up/down) each contribute a centered second difference:

The two-dimensional wave equation (cont.)



four centered differences
inside the circular rim

The two-dimensional wave equation (cont.)

$$\frac{d^2 u_{ij}}{dt^2} = c^2 \left[\frac{u_{i+1,j} - 2u_{ij} + u_{i-1,j}}{(\delta x)^2} + \frac{u_{i,j+1} - 2u_{ij} + u_{i,j-1}}{(\delta y)^2} \right], \qquad (1.2.1)$$

Here c is the physical wave speed of the membrane. If the two lattice directions used different spacings or spring scalings, the limiting equation would carry different constants in the x and y directions. In the isotropic limit $\delta x, \delta y \to 0$, each bracketed term becomes a second partial derivative,

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right]. \qquad (1.2.2)$$

Writing ∇^2 for the Laplacian (the sum of the unmixed second derivatives), this is more succinctly

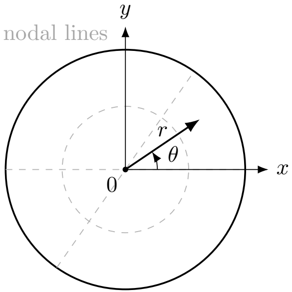
$$\frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} = \nabla^2 u. \qquad (1.2.3)$$

The time derivative is *second* order: the drum, like the string, obeys a genuine wave equation, not a diffusion equation.

The two-dimensional wave equation (cont.)

The boundary of the drum is a circle, so polar coordinates (r, θ) make the boundary condition simple. Away from the coordinate singularity at $r = 0$, the Laplacian is

$$\nabla^2 u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\,\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}. \tag{1.2.4}$$



rim: $u(1, \theta, t) = 0$

The two-dimensional wave equation (cont.)

Expanding the chain-rule calculation skipped in the book, since $x = r\cos\theta$ and $y = r\sin\theta$,

$$\partial_x = \cos\theta\,\partial_r - \frac{\sin\theta}{r}\,\partial_\theta, \qquad \partial_y = \sin\theta\,\partial_r + \frac{\cos\theta}{r}\,\partial_\theta.$$

Applying these two operators twice and adding, the mixed $u_{r\theta}$ terms cancel and the remaining coefficient derivatives combine to $u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}$, which is exactly (1.2.4). Inserting (1.2.4) into (1.2.3) gives the polar wave equation for $u(r,\theta,t)$:

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}. \tag{1.2.5}$$

The two-dimensional wave equation (cont.)

For a drum head of radius 1, idealized with prescribed initial shape $u_0(r,\theta)$ and zero initial velocity, the conditions are

$$\begin{aligned}
\text{initial:}\quad & u(r,\theta,0) = u_0(r,\theta), \qquad \tfrac{\partial}{\partial t} u(r,\theta,0) = 0,\\
\text{boundary:}\quad & u(1,\theta,t) = 0, \qquad u(r,0,t) = u(r,2\pi,t),\\
& \tfrac{\partial}{\partial \theta} u(r,0,t) = \tfrac{\partial}{\partial \theta} u(r,2\pi,t),\\
\text{center:}\quad & u(0,\theta,t) \text{ is finite and independent of } \theta.
\end{aligned}$$

The rim is clamped ($r = 1$ at the edge gives zero displacement there), the center condition is regularity at the coordinate singularity $r = 0$, and the periodic lines say the solution is single-valued as θ wraps around the circle: $\theta = 0$ and $\theta = 2\pi$ are the same physical point. For a sharply struck drum, the complementary idealization is also common: one takes $u(r,\theta,0) = 0$ and prescribes a nonzero initial velocity $\partial_t u(r,\theta,0) = v_0(r,\theta)$.

Separation in time: the Helmholtz equation

As before, look for a separable solution—now the time part splits off first:

$$u(r,\theta,t) = H(r,\theta)\,T(t). \tag{1.2.6}$$

Substituting into the polar wave equation (1.2.5) gives $\frac{1}{c^2}HT'' = T\,\nabla^2 H$. Dividing by HT,

$$\frac{1}{c^2}\,\frac{T''}{T} = \frac{\nabla^2 H}{H}. \tag{1.2.7}$$

The left side depends only on t; the right only on (r,θ). Both must equal one constant λ. For a nonzero clamped mode, λ must be negative: multiplying $\nabla^2 H = \lambda H$ by H and integrating over the disk gives $-\int |\nabla H|^2\,dA = \lambda \int H^2\,dA$, because the boundary term vanishes at the clamped rim. Thus set $\lambda = -k^2$ with $k > 0$. This yields two equations:

$$T'' = -c^2 k^2\,T, \tag{1.2.8}$$

$$\nabla^2 H = -k^2 H. \tag{1.2.9}$$

The time equation (1.2.8) solves to $T(t) = C\cos ckt + D\sin ckt$. The release-from-rest condition $\partial_t u = 0$ at $t = 0$ forces $D = 0$ (since $T'(0) = ckD$), leaving

$$T(t) = A\cos ckt. \tag{1.2.10}$$

Note the time frequency is ck, set by the wave speed times the separation constant.

Separation in time: the Helmholtz equation (cont.)

Equation (1.2.9) is the *Helmholtz equation*, usually written

$$\nabla^2 H + k^2 H = 0. \tag{1.2.11}$$

In polar coordinates, using (1.2.4),

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial H}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 H}{\partial \theta^2} + k^2 H = 0. \tag{1.2.12}$$

We separate the spatial part once more, $H(r,\theta) = R(r)\,\Theta(\theta)$. Substituting and writing $(rR')' = \frac{d}{dr}(rR')$,

$$\Theta\,\frac{1}{r}(rR')' + R\,\frac{1}{r^2}\Theta'' + k^2 R\Theta = 0.$$

Multiply through by $r^2/(R\Theta)$:

$$\frac{r(rR')'}{R} + \frac{\Theta''}{\Theta} + k^2 r^2 = 0,$$

and move the θ-term to the other side,

$$r\,\frac{(rR')'}{R} + k^2 r^2 = -\frac{\Theta''}{\Theta}. \tag{1.2.13}$$

Separation in time: the Helmholtz equation (cont.)

The left side is a function of r only, the right of θ only, so both equal a constant μ:

$$r\,\frac{(rR')'}{R} + k^2 r^2 = -\frac{\Theta''}{\Theta} = \mu. \tag{1.2.14}$$

This gives the angular and radial equations

$$r\,(rR')' + (k^2 r^2 - \mu)R = 0, \tag{1.2.15}$$

$$\Theta'' + \mu\,\Theta = 0. \tag{1.2.16}$$

The angular equation and periodicity

Start with the angular part (1.2.16), $\Theta'' + \mu\Theta = 0$. If $\mu < 0$, the solutions are hyperbolic/exponential and cannot be 2π-periodic unless they are trivial. For $\mu = 0$ the solution is $\Theta = A + B\theta$, and periodicity forces $B = 0$, leaving the constant angular mode. For a genuine oscillation around the rim we need $\mu > 0$; write the solution as

$$\Theta(\theta) = A\cos\!\big(\sqrt{\mu}\,\theta\big) + B\sin\!\big(\sqrt{\mu}\,\theta\big), \tag{1.2.17}$$

for coefficients A and B. The drum head is one connected surface, so Θ must return to its starting value after a full turn: $\Theta(0) = \Theta(2\pi)$ (and likewise for its derivative). With $\Theta(0) = A$ and $\Theta(2\pi) = A\cos(2\pi\sqrt{\mu}) + B\sin(2\pi\sqrt{\mu})$, and the matching $\Theta'(0) = \sqrt{\mu}\,B$ against $\Theta'(2\pi) = \sqrt{\mu}\big(-A\sin(2\pi\sqrt{\mu}) + B\cos(2\pi\sqrt{\mu})\big)$, both conditions hold for arbitrary A, B exactly when $\cos(2\pi\sqrt{\mu}) = 1$ and $\sin(2\pi\sqrt{\mu}) = 0$, i.e. when $\sqrt{\mu}$ is a whole number. Hence

$$\sqrt{\mu} = m \quad \text{for some nonnegative integer } m, \qquad \text{i.e.} \qquad \mu = m^2. \tag{1.2.18}$$

The integer m counts how many full angular oscillations fit around the drum; $m = 0$ is the radially symmetric mode.

The angular equation and periodicity (cont.)

Substituting $\mu = m^2$ into the radial equation (1.2.15),

$$r\,(rR')' + (k^2 r^2 - m^2)R = 0. \tag{1.2.19}$$

The factor k^2 is a nuisance; remove it by rescaling the radius. Introduce the dimensionless radius $s = kr$ and let $\widetilde{R}(s) = R(r)$ be the same displacement read off against s. By the chain rule each r-derivative brings down a factor k,

$$\frac{dR}{dr} = k\,\frac{d\widetilde{R}}{ds}, \qquad \frac{d^2R}{dr^2} = k^2\,\frac{d^2\widetilde{R}}{ds^2},$$

and $r = s/k$, so $k^2r^2 = s^2$. Substituting into the expanded form $r^2R'' + rR' + (k^2r^2 - m^2)R = 0$ of (1.2.19),

$$\frac{s^2}{k^2}\,k^2\widetilde{R}'' + \frac{s}{k}\,k\widetilde{R}' + (s^2 - m^2)\widetilde{R} = 0,$$

and the powers of k cancel termwise. Renaming s back to r and $\widetilde{R}$ to R, the rescaled equation is

$$r\,(rR')' + (r^2 - m^2)R = 0. \tag{1.2.20}$$

The angular equation and periodicity (cont.)

Expanding the derivative $r(rR')' = r(R' + rR'') = r^2R'' + rR'$ gives the standard form

$$r^2R'' + rR' + (r^2 - m^2)R = 0. \tag{1.2.21}$$

This is *Bessel's equation* of order m.

Bessel's equation and the drum solution

The solution of Bessel's equation (1.2.21) cannot be written as a finite product and sum of elementary functions. The two independent radial solutions are traditionally called $J_m(r)$ and $Y_m(r)$. The function Y_m is singular at the center (Y_0 has a logarithmic singularity, and Y_m for $m \geq 1$ blows up like r^{-m}), so the boundedness and smoothness condition at $r = 0$ removes it. The regular solution is the *Bessel function* $J_m(r)$, normalized by $J_m(r) \sim (r/2)^m/m!$ as $r \to 0$. Naming it lets us write the answer in closed form even though no elementary formula exists.

Assembling the pieces—radial factor $J_m(kr)$ (undoing the rescaling), angular factors $\cos(m\theta)$ and $\sin(m\theta)$, and time factor $\cos(ckt)$ from (1.2.10)—and superposing over all admissible angular integers m and all allowed radial constants k, the displacement of the drum is

$$u(r,\theta,t) = \sum_{m=0}^{\infty}\sum_{j=1}^{\infty}\Big[A_{jm}\cos(m\theta) + B_{jm}\sin(m\theta)\Big]\, J_m(k_{jm}r)\,\cos(ck_{jm}t). \tag{1.2.22}$$

Bessel's equation and the drum solution (cont.)

For $m = 0$ the sine term is zero and may be dropped. The allowed values $k_{jm} = \alpha_{m,j}$ are the positive zeros of J_m, so that the rim condition $u(1, \theta, t) = 0$ holds. The radial orthogonality used to find the coefficients is

$$\int_0^1 r\, J_m(\alpha_{m,j} r) J_m(\alpha_{m,\ell} r)\, dr = \frac{1}{2} J_{m+1}^2(\alpha_{m,j})\, \delta_{j\ell}.$$

Together with the usual sine/cosine angular orthogonality, this fixes A_{jm} and B_{jm} from the initial shape, exactly as Fourier coefficients were fixed for the string.

Bessel's equation and the drum solution (cont.)

Writing

$$N_{mj} := \int_0^1 rJ_m^2(\alpha_{m,j}r)\,dr = \frac{1}{2}J_{m+1}^2(\alpha_{m,j}),$$

the projection formula for $m \geq 1$ is

$$A_{jm} = \frac{1}{\pi N_{mj}} \int_0^{2\pi}\int_0^1 u_0(r,\theta)J_m(\alpha_{m,j}r)\cos(m\theta)\,r\,dr\,d\theta,$$

and B_{jm} is the same formula with $\sin(m\theta)$. For $m = 0$, the angular norm is 2π instead of π, and the sine coefficient is absent:

$$A_{j0} = \frac{1}{2\pi N_{0j}} \int_0^{2\pi}\int_0^1 u_0(r,\theta)J_0(\alpha_{0,j}r)\,r\,dr\,d\theta, \qquad B_{j0} = 0.$$

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Naming the functions: the road ahead

The shape of the boundary decides which functions appear.

- **Guitar string** (interval): the modes are sinusoids; the expansion is a *Fourier* series.
- **Circular drum**: the radial modes are *Bessel functions* J_m.
- **Rectangular drum**: the solution would again be built from sinusoidal Fourier modes, like the string.
- **Elliptical drum**: the solutions are *Mathieu functions*.

Many differential equations of physics cannot be solved with elementary functions, but their solutions arise so often that we assign them proper names. This class of named solutions is the family of *special functions*.

The purpose of the course is to study several of the most important special functions—the Bessel function, the gamma function and its relatives, and the orthogonal polynomials. To analyze them comfortably we will first develop some tools of complex analysis.